

FiTech GA5 Project Report

1. Problem Statement & Solution

Gym users frequently encounter physical and cognitive obstacles when using traditional fitness apps during their workout. For experienced lifters, wearing lifting straps or having sweaty hands makes touch-screen interaction physically difficult. Beginners often struggle with digital distractions—such as mindless scrolling during rest periods—and face cognitive overload when a crowded gym forces them to find alternative equipment.

FiTech is a screen-less, voice-first fitness assistant designed to solve these problems, eliminating physical interaction with phone and preventing digital distractions.

2. Core Tasks

The system supports several essential tasks to ensure a screen-less workout experience:

- **Real-time Audio Guidance:** The assistant delivers immediate instructions through wireless earbuds regarding the next exercise, target weight, reps, and remaining rest intervals.
- **Low-Interaction Data Logging and Control:** User can record workout data with no screen-interaction. Users can control the session using simple earbud taps to do different tasks.
- **Post-Workout Analytics & Progression Tracking:** After the session, the app transforms collected data into visual insights, charting milestones such as progressive overload.

3. Specific Implementation

- **Voice-First Interface:** An audio-based interaction model enables an "eyes-free" experience, removing the need for visual menu navigation and preventing digital context switching.
- **AI Adaptive Engine:** The system generates optimized workout routines based on user-defined inputs, such as target muscle groups and workout duration, while providing the flexibility for users to manually modify and customize the plan to suit their specific preferences (e.g. weight, set, time).
- **Integrated Physical Controls:** Simple hardware interactions are utilized as event triggers to minimize physical effort.
 - **Single Tap:** Advances the user to the next session (confirming set completion)
 - **Double Tap:** Skips the current exercise. ("I do not want to do this exercise today.")
 - **Triple Tap:** Indicates that the equipment is occupied, moving the current exercise to the next order. ("There are people using the machine I have to use. I will have to instead do this one after the next exercise.")